Retrieval that Cannot See the Future: Temporal Evaluation of Fact-Checked Claim Retrieval Project Proposal for NLP Course, Winter 2026/27

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Abstract

Previously fact-checked claim retrieval (PFCR) ranks an archive of fact-checks for a social media post. Published benchmarks search the full archive, including fact-checks written after the post, while a fact-checker at post time can only search what already exists. We propose a *causal* evaluation protocol: for each post q with time t_q, the pool is $P(q) = \{f : t_f < t_q\}$, and we report Success@10 (fraction of queries with a gold fact-check in the top 10) and mean reciprocal rank (MRR) under random, size-matched random, and strict-causal pools on MultiClaim, with Polish as focus and English as control. Deliverables: the protocol and rebuildable split as id lists and code; a results table of at least four systems under both protocols; and a gap decomposition with 50 hand-coded newly failing queries.

1 Scientific goal

Problem. PFCR matches a post q to earlier fact-checks f that already debunk the same claim (Shaar et al., 2020; Pikuliak et al., 2023). Standard evaluation lets q retrieve fact-checks published after t_q, which rewards matching a later article that quotes the post. We ask how much reported performance survives a time-respecting index, and why it drops.

Research questions. RQ1: How much Success@10 (S@10) and mean reciprocal rank (MRR) survive a causal index (unconditionally and on answerable queries), and does system ranking change? **RQ2:** How much of the random-vs-causal gap is answerability loss, pool-size effects, training leakage, or missing labels for earlier sibling debunks? **RQ3:** Is the causal penalty larger for Polish than for English, and how much do earlier English fact-checks recover for Polish posts? **RQ4:** Do large language model (LLM) rerankers gain on facts they may have seen in pretraining (core+, after the committed core)?

Hypotheses (pre-registered). Most MultiClaim gold pairs are post-before-fact-check (H1); on answerable queries, time-local distractors are harder than a size-matched random pool (H2); Polish shows a larger causal penalty than English, especially for lexical BM25 (H3); fine-tuned retrievers lose more under chronological training than zero-shot ones (Shaar et al., 2022) (H4).

2 Significance and state of the art

We summarise fifteen prior works. Table 1 lists the published numbers we compare, with the dataset and split named for every value. No PFCR evaluation we found restricts the fact-check index *per query* to documents published before the post.

PFCR task and shared tasks. (Shaar et al., 2020) define PFCR as ranking verified claims for a social-media or debate input; on Snopes tweets and PolitiFact debate sentences, BM25 plus rankSVM reaches MRR 0.788 and S@10 0.915 on the Snopes test (BM25 top-50 rerank). (Shaar et al., 2021) run CLEF CheckThat! 2021 Task 2 (English/Arabic tweets; English debates) with fixed pools and no date filter; best English tweet MAP@5 is 0.883 (Aschern) vs BM25 0.749 on the 70/15/15 tweet split. (Nakov et al., 2022) rerun the lab in 2022 with a small new test set; RIET Lab reaches MAP@5 0.956 on English

tweets (2022 test) with sentence-T5 plus GPT-Neo reranking. (Hardalov et al., 2022) mine 333k noisy tweet–Snopes pairs from crowd replies and train sentence-BERT; on CheckThat! 2021 English test they report MAP@5 90.3 for the full system (vs Aschern 88.2).

Multilingual PFCR corpora and systems. (Pikuliak et al., 2023) release MultiClaim (28k posts, 206k fact-checks, 31k pairs) and evaluate unsupervised and fine-tuned retrievers over the *full* pool; best monolingual S@10 is 0.83 (GTR-T5-Large on English translations). Their post-newer vs post-older subsets score 80.02 vs 80.04 S@10 under the same full index (no per-query date filter). (Peng et al., 2025) organise SemEval-2025 Task 7 on MultiClaim with out-of-time test posts (pool to May 2024); winner PINGAN AI reaches monolingual average S@10 0.9601 and Polish 0.9260 on the hidden test languages. (Devadiga et al., 2025) (SemEval system) translate posts, fine-tune Stella with hard negatives and LLM-rerank; monolingual test S@10 0.938 (Polish 0.886). (Ramponi et al., 2025) compare multilingual vs crosslingual retrieval on a curated MultiClaim v2 subset (47 languages) with an 80/10/10 language-stratified split that separates fact-checks across folds; LLM reranking of multilingual-e5-large reaches MRR@10 0.8042 multilingual. (Vykopal et al., 2025) release AMC-16K (16k human Yes/No pairs from MultiClaim; 16% relevant overall) and evaluate LLMs as relevance detectors; best Macro-F1 82.46 monolingual (Mistral Large few-shot).

Closest temporal precedents for PFCR. (Shaar et al., 2022) add transcript context and a debate-level chronological train/test split on PolitiFact; the prior SOTA reranker drops from MAP 0.602 (sentence-level random) to 0.429 (debate chrono) without restricting the claim pool by date.

Time-respecting IR and verification (not PFCR index filtering). (Ounis et al., 2011) define TREC 2011 Microblog real-time search with a compulsory run that bans future tweets and future corpus statistics; best compulsory P@30 is 0.4551. (Liška et al., 2022) evaluate QA over a growing dated news corpus by quarter; dated DPR raises recall@20 on recent generated questions from 43% to 57%. (Glockner et al., 2022) show that 69.7% of MultiFC misinformation claims have leaked fact-check evidence in search snippets (verification leakage from later fact-checks). (Schlichtkrull et al., 2023) release AVeriTeC with evidence restricted to documents before the claim and temporally ordered splits; baseline verdict macro-F1 is 0.23 (gold evidence 0.49) on the AVeriTeC test. (Mu et al., 2023) show rumour classifiers collapse under chronological splits: on Twitter 16, BERT macro-F1 falls from 91.5 (random) to 47.2 (stratified chronological).

Table metrics. Let $G(q)$ be the gold items for query q and r_q the rank of the first gold hit (undefined if none). *S@10* is the fraction of queries with a gold hit in the top 10; *MRR* is the mean of $1/r_q$ (0 if no gold ranked) (Pikuliak et al., 2023; Peng et al., 2025). *MRR@10* clips reciprocal rank at 10. *MAP@k* averages, over queries, precision at each relevant hit within the top k (plain MAP uses the full list). *P@k / R@k*: fraction of the top k that are relevant / fraction of all gold recovered in the top k. *Macro-F1*: unweighted mean of per-class F1 (classification, not ranking). *Leak rate*: share of claims whose later fact-check evidence already appears in search snippets (Glockner et al., 2022).

3 Datasets and status

Table 2 lists the source datasets (derived annotations such as failure codes are omitted here). For ranking corpora, class balance is the density of positive post–fact-check links, not a balanced classification split.

4 Concept, methodology and work plan

Pipeline. Figure 1 summarises the evaluation path. Systems (committed core): BM25; multilingual-e5-large (Wang et al., 2024); a Polish or cross-lingual encoder (mmlw / LaBSE); bge-reranker-v2-m3 on the top 100 (Chen et al., 2024). Protocols: full-pool random; strict causal $P(q)$; size-matched random of $|P(q)|$. Compute: local open-weight models only; no commercial LLM API on post text.

Metrics. Chosen before results (MultiClaim / SemEval-2025 convention (Pikuliak et al., 2023; Peng et al., 2025)). Let $G(q)$ be the gold fact-checks for query q and r_q the rank of the first gold hit ($1/r_q = 0$ if none). *S@10* is the fraction of queries with $r_q \leq 10$; *MRR* is the mean of $1/r_q$. *Answerable rate* is the share of queries with $G(q) \cap P(q) \neq \emptyset$ under the causal pool. We report

Table 1: Published numbers for the fifteen works above. Every value names the dataset and the split or evaluation condition used in the source paper. Metrics as above.

Work	Metric	Value	Dataset / split
(Shaar et al., 2020)	MRR	0.788	Snopes tweets / official test; BM25 top-50 + rankSVM
(Shaar et al., 2020)	S@10	0.915	same
(Shaar et al., 2021)	MAP@5	0.883	CheckThat! 2021 EN 2A / 70-15-15 test (Aschern)
(Shaar et al., 2021)	MAP@5	0.749	same / BM25 baseline
(Nakov et al., 2022)	MAP@5	0.956	CheckThat! 2022 EN 2A / 2022 test (RIET Lab)
(Hardalov et al., 2022)	MAP@5	90.3	CheckThat! 2021 EN test / CrowdChecked-trained system
(Pikuliak et al., 2023)	S@10	0.83	MultiClaim / full pool, mono, GTR-T5-L (EN text)
(Pikuliak et al., 2023)	S@10	80.02 / 80.04	MultiClaim / post-newer vs older, full index, GTR-T5
(Peng et al., 2025)	S@10	0.9601	SemEval-2025 / mono test avg (PINGAN AI)
(Peng et al., 2025)	S@10	0.926	SemEval-2025 / Polish test (PINGAN AI)
(Devadiga et al., 2025)	S@10	0.938	SemEval-2025 / mono test (TIFIN)
(Devadiga et al., 2025)	S@10	0.886	SemEval-2025 / Polish test (TIFIN)
(Ramponi et al., 2025)	MRR@10	0.804	MultiClaim v2 curated / 80-10-10 + LLM rerank
(Vykopal et al., 2025)	Macro-F1	82.46	AMC-16K / mono human labels, Mistral few-shot
(Shaar et al., 2022)	MAP	0.602	PolitiFact debates / sentence-level random
(Shaar et al., 2022)	MAP	0.429	PolitiFact debates / debate-level chronological
(Ounis et al., 2011)	P@30	0.455	TREC 2011 Microblog / compulsory no-future run
(Liška et al., 2022)	R@20	57% / 43%	StreamingQA / 2020 recent gen. Qs; dated vs undated DPR
(Glockner et al., 2022)	leak rate	69.7%	MultiFC misinfo claims / auto leak detector
(Schlichtkrull et al., 2023)	macro-F1	0.23 / 0.49	AVeriTeC test / baseline vs gold evidence
(Mu et al., 2023)	macro-F1	91.5 / 47.2	Twitter 16 / random vs stratified chronological, BERT

both unconditionally (unanswerable queries count as misses) and on answerable queries only; significance via paired cluster bootstrap; Polish–English contrast via difference-in-differences.

Concrete success deliverables. (1) Rebuildable causal split (id lists + code; no post text). (2) Shared results table: $\geq$4 systems $\times$ random vs causal $\times$ Polish and English, with bootstrap intervals; central number = extra Polish temporal cost. (3) Gap decomposition plus coded analysis of 50 newly failing queries.

Week-by-week plan. Internal work finishes before Christmas; course checkpoints: mid-term video 18 Nov, Phase 2 2 Dec, formal final 13 Jan 2027.

- **W1 (to 18 Nov):** download MultiClaim v1/v2 and SemEval-2025 Task 7, dated corpus, protocol module, answerable-rate gate. Owner: Paweł Pozorski.
- **W2 (to 2 Dec):** core systems (BM25, dense, reranker) for Polish and English; Phase 2 PoC. Owner: Mateusz Jarosz (with Paweł Florek on GPU runs).
- **W3 (to 16 Dec):** bootstrap tables, gap decomposition, 50 failure annotations. Owner: Paweł Florek (with Jan Cwalina on annotation).
- **W4 (to 23 Dec):** release check, write-up; core+ only if time remains. Project deliverables done before Christmas. Owner: Jan Cwalina (with Paweł Pozorski on writing).

Main risks: unknown Polish answerable rate (if $<$150, Polish–English contrast becomes descriptive); timestamp semantics undocumented (URL date audit + margin sensitivity); AMC-16K still to be requested.

5 Data statement

Primary data: MultiClaim v1/v2 and SemEval-2025 Task 7 from restricted Zenodo records (research use only; no re-identification or re-sharing beyond named requesters) (Moro et al., 2025; Pikuliak et al., 2023; Peng et al., 2025); MultiClaimNet (Zenodo; CC BY 4.0) (Panchendrarajan et al., 2025). Access is granted for MultiClaim v1/v2, SemEval-2025 Task 7 and MultiClaimNet. Posts are real social-media text, released in anonymised form by the dataset authors; we treat them as personal data. We store files only in a directory readable by named team members, never send post text to an external LLM API, never put raw text in git or the paper (examples paraphrased; errors aggregated), and delete the data at project end. AMC-16K is secondary (LLM-judge validation only). Its public listing is on the EU

Table 2: Source datasets. FC = fact-checks: short verified/debunked claims from fact-checking sites (the documents we retrieve).

Dataset	Size	Label definition	Class balance	Licence	How obtained	Status
MultiClaim v1	206k FC, 28k posts, 31k pairs	fact-checker gold links (ClaimReview / warnings)	sparse positives (ranking)	research use only	Zenodo 7737983	READY
MultiClaim v2	435k FC, 89k posts, 105k pairs	same + relationship types	sparse positives (ranking)	research use only	Zenodo 15413169	READY
SemEval-2025 T7	train from MultiClaim; test 4.3k mono posts (PL: 500 / 564 pairs)	shared-task gold for S@10	sparse positives (ranking)	research use only	Zenodo 14989176	READY
MultiClaimNet	85.3k FC in 30.9k clusters	same-claim cluster ids	n/a (clustering)	CC BY 4.0	Zenodo 15100352	READY
AMC-16K	16k pairs / 20 langs	human Yes/No relevance	16% relevant overall; PL mono 20.25%	CC BY-NC-SA 4.0	authors (not via LDS)	RESTRICTED – to be requested

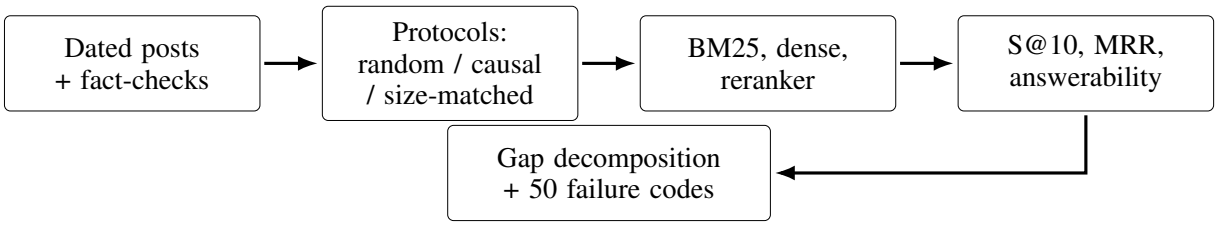


Figure 1: Evaluation pipeline.

Language Data Space catalogue, which WUT cannot use; we will request a copy from the authors (KInIT) under research terms.

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